TOBB ETÜ

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BIL 496 TEST PLAN REPORT

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1. Introduction

This test plan describes how the AI-Assisted Ultrasound Diagnosis System will be tested. The aim is to verify that the system works correctly, meets all project requirements, and produces expected results in both the Doctor and Admin workflows.
The system enables doctors to query patient ultrasound examinations, select diagnostic images across six anatomical regions (R1–R6), apply configurable preprocessing operations, submit images to an AI diagnostic module, and generate exportable reports. Admin users have additional capabilities to manage user accounts.

2. Project Scope

Project Title: NeoAI: AI-Assisted Diagnosis System

Brief Description: A medical software platform that allows doctors to query patient health data with AI-driven diagnostics, and produce clinical reports.

Main Functionalities:

- User authentication (Doctor and Admin roles)
- Patient data and examination querying by ID
- Ultrasound video browsing and image selection (R1–R6)
- Configurable preprocessing pipeline
- AI model selection and diagnostic processing
- Report generation with commentary and export (PDF / DOC)
- Admin user management and audit logging

3. Features to Be Tested

3.1 Doctor Workflow

- Login and session management
- Patient data query by ID
- Examination selection and video browsing
- Image selection: exactly one per region (R1–R6)
- Preprocessing operation selection and configuration
- AI model selection
- Preprocessing execution and AI inference
- Report viewing, commentary, and export (PDF & DOC)
- Preprocessed data and audit record persistence

3.2 Admin Workflow

- All Doctor features
- Admin Panel access (restricted to admin role)
- New user creation (no self-registration)
- User settings modification
- Audit record creation for admin actions

4. Testing Methodology

The following testing methods are applied in this project:

Unit Testing: Testing individual functions or modules such as the preprocessing engine, AI interface, and report generator in isolation.

Integration Testing: Testing interactions between modules — for example, verifying that the preprocessing pipeline correctly feeds processed images to the AI module.

System Testing: End-to-end testing of the complete system covering full workflows for both Doctor and Admin roles.

Security & Role Testing: Verifying that role-based access controls are enforced: Doctors cannot access the Admin Panel; unauthenticated users cannot access any protected screens.

Validation Testing: Verifying that required constraints are enforced, such as selecting exactly one image per region before proceeding, and mandatory AI model selection.

5. Test Environment

Hardware

- Personal computer (development machine)
- Standard CPU-based system (no dedicated GPU required)
- Network connection for AI module communication

Software

- Operating System: Windows / Linux
- Programming Language: Java
- Framework: Spring Boot
- Database: PostgreSQL / SQL Database
- REST API communication tools (Postman or similar)

Test Data: De-identified sample patient records including ultrasound video data with at least one examination covering all six anatomical regions (R1–R6).

6. Test Cases

Each test case is identified by a unique Test ID. The table below covers all major features of both the Doctor and Admin workflows.

Test ID	Description	Input(s)	Expected Output	Result
TC-01	Doctor login with valid credentials	Valid username & password	Login successful; main menu displayed	
TC-02	Doctor login with invalid credentials	Wrong password	Error message shown; access denied	
TC-03	Admin login with valid credentials	Admin username & password	Login successful; main menu + Admin Panel available	
TC-04	Unauthorized access attempt without login	Direct URL access	Redirect to login page	
TC-05	Query patient data by valid patient ID	Existing patient ID	List of examinations with ID and date shown	
TC-06	Query patient data by non-existent ID	Non-existent patient ID	Appropriate 'no records found' message	
TC-07	Select an examination from the list	Click on examination entry	Videos for all regions (R1-R6) displayed	
TC-08	Select one image per region (R1-R6)	Select 1 image from each region video	6 images selected; approve button enabled	
TC-09	Attempt to proceed without selecting all regions	Select images for only 4 of 6 regions	Warning message; progression blocked	
TC-10	Approve image selection	All 6 regions selected, click approve	Preprocessing panel opened	
TC-11	Load preprocessing settings panel	Approved image selection	Preprocessing operations panel displayed	
TC-12	Apply preprocessing operation (e.g., noise reduction)	Select noise reduction, set intensity=5	Operation listed with parameters	
TC-13	Change preprocessing operation settings	Modify existing operation parameter	Updated value reflected in settings panel	
TC-14	Remove a preprocessing operation	Deselect an applied operation	Operation removed from list	

Test ID	Description	Input(s)	Expected Output	Result
TC-15	Select AI model for diagnosis	Choose available AI model from list	AI model selected; ready to process	
TC-16	Proceed without selecting an AI model	Click continue with no AI selected	Error prompt; progression blocked	
TC-17	System applies preprocessing and sends to AI	Valid images + operations + AI model	Processing completes; report generated	
TC-18	View generated diagnostic report	Completed processing	Report panel displayed with AI results	
TC-19	Add commentary to the report	Type commentary text in report panel	Commentary saved and visible in report	
TC-20	Export report as PDF	Click export, select PDF	PDF file downloaded successfully	
TC-21	Export report as DOC	Click export, select DOC	DOC file downloaded successfully	
TC-22	Verify preprocessed data is saved with operations	Complete a full diagnosis workflow	Data + operations stored in system	
TC-23	Verify audit record created for doctor action	Complete a diagnosis session	Audit log entry with user, action, timestamp	
TC-24	Navigate back to data query from report panel	Click back/return in report panel	Return to patient data query screen	
TC-25	Admin accesses Admin Panel	Admin login, click 'Admin Panel'	Admin panel displayed	
TC-26	Admin creates a new user	Fill in new user details, confirm	New user account created; audit record stored	
TC-27	Admin modifies existing user settings	Select user, change settings, approve	Settings updated; audit record stored	
TC-28	Non-admin user tries to access Admin Panel	Doctor account attempts admin access	Access denied; panel not shown	
TC-29	Admin creates duplicate username	Enter existing username in new user form	Error message; user creation blocked	
TC-30	Audit record captures admin identity on user creation	Admin creates a new user	Audit log shows admin name, action, timestamp	

7. Risks and Limitations

Possible Technical Risks:

- AI model inference may return inconsistent results for low-quality or edge-case ultrasound images.
- Preprocessing operations with extreme parameter values may distort images before AI submission.
- Report generation failures if the AI module returns an unexpected or malformed response.
- Concurrent access by multiple users may cause session or data consistency issues.

Dataset and Hardware Limitations:

- Testing is limited to the available de-identified sample dataset; results may not generalize to all real-world ultrasound data.
- AI diagnostic accuracy is dependent on the training data and model version used during testing.
- AI inference testing may be significantly slower and may not reflect production performance.

8. Conclusion

This test plan covers all critical functionalities of the AI-Assisted Ultrasound Diagnosis System for both Doctor and Admin roles. The 30 defined test cases span authentication, data querying, image selection, preprocessing, AI diagnosis, report generation, data persistence, audit logging, and role-based access control.

All test cases are expected to pass under normal operating conditions with properly configured hardware, software, and test data. Any failures identified during testing should be documented, triaged, and resolved before the system is considered ready for production use.

Req. ID	Requirement Description	Related Test Cases	Status
Functional Requirements			
FR1	Ultrasound data ingestion from connected DB	TC-05, TC-06, TC-07	**Covered**
FR2	Clinician viewing, processing & analysis of ultrasound data	TC-07, TC-08, TC-09, TC-10, TC-11, TC-12, TC-13, TC-14, TC-15, TC-16, TC-17, TC-18, TC-19, TC-20, TC-21	**Covered**
FR3	Store processing steps for traceability	TC-22, TC-23	**Covered**
FR4	Transmit processed ultrasound data to AI module	TC-17	**Covered**
FR5	Receive & store AI diagnosis results	TC-17, TC-18	**Covered**

Req. ID	Requirement Description	Related Test Cases	Status
FR6	Create/maintain AI training datasets from ultrasound data	TC-22	**Covered**
FR7	User authentication (login)	TC-01, TC-02, TC-03, TC-04	**Covered**
FR8	Role-based access control	TC-03, TC-04, TC-25, TC-28	**Covered**
FR9	Data anonymization / pseudonymization	TC-09	**Covered**
FR10	Immutable audit log of all user actions	TC-23, TC-24, TC-30	**Covered**
Non-Functional Requirements			
NFR1	System responsiveness	TC-17, TC-18	**Covered**
NFR2	Ease of use for non-technical users	TC-08, TC-10, TC-11, TC-18, TC-19	**Covered**
NFR3	24/7 availability	TC-01, TC-17	**Covered**

NOTE: Following discussions with the company, the data type requirement has been changed from oximeter to ultrasound. All requirements are the same.This information is provided for your awareness.